

UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy
DEGREE OF M.Sci. WITH HONOURS
FOURTH-YEAR FINAL EXAMINATION

03 30682

LM RELATIVISTIC ASTROPHYSICS

SEMESTER 2 EXAMINATIONS 2021/22

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets [].

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and two tables of physical constants and units that may be required will be found at the end of this question paper.

ANY CALCULATOR

Answer **two** questions. If you answer more than two questions, credit will only be given for the best two answers.

1. Assume that a neutron star is composed of a zero-temperature gas of N ultra-relativistic neutrons where the total Fermi energy of the neutrons equals the gravitational potential energy of the star.

(a) Provide an order of magnitude estimate of the total number of neutrons, N . **[8]**

(b) Hence, estimate the mass of the star. **[3]**

(c) A neutron star of mass $m = 1.3 M_{\odot}$ and radius $R = 12$ km is captured by a black hole of mass $M = 6 M_{\odot}$.

To tackle this problem work at the leading Newtonian order.

i. Estimate the separation at which the star is tidally disrupted by the black hole **[7]**

ii. Hence, discuss whether photons and neutrinos produced during the shredding of the star would be in principle observable. **[3]**

iii. If the neutron star plunges into the black hole starting from rest at a large distance, $r \gg 1$, estimate the total energy that is radiated in gravitational waves during this process. Energy is radiated according to

$$\frac{dE}{dr} = \frac{8\sqrt{2}}{15}(mM)^2 \sqrt{(m+M)r^{-9/2}}, \quad [c = G = 1],$$

and r is the neutron star-black hole radial distance. **[9]**

2. The double pulsar is currently the only known binary system in which both neutron stars are observed as radio pulsars (named pulsar A and pulsar B).

Pulsar A has a rotation frequency $\nu_A = 44 \text{ Hz}$. The first and second time derivative of the frequency have been measured to be $\dot{\nu} = -3.4 \times 10^{-15} \text{ Hz s}^{-1}$ and $\ddot{\nu} = -9.2 \times 10^{-31} \text{ Hz s}^{-2}$, respectively. The moment of inertia, I_A , of this neutron star has also been measured, and its value is $I_A = 1.3 \times 10^{45} \text{ gr cm}^2$.

(a) Estimate the spin-down age of pulsar A. **[2]**

(b) Estimate the rate of rotational energy loss of pulsar A. **[6]**

(c) How long would it take for the neutron star to lose 10% of its rotational energy? **[4]**

(d) In the simple magnetic dipole model of radio pulsars the star rotational energy is radiated away due to the time varying dipole moment associated to the magnetic field, B . In this model, the energy is lost at a rate

$$\frac{dE}{dt} = -\frac{B^2 R^6 (2\pi\nu)^4 \sin^2 \alpha}{6c^3},$$

where B is the neutron star's magnetic field, R the radius of the neutron star, and α the angle between the magnetic field and the star's rotation axis.

i. Derive a constraint on the magnetic field of pulsar A. **[9]**

ii. From the result that you obtain discuss whether you think this simple model provides a satisfactory physical description of pulsar A. **[3]**

iii. Use the measured spin-down parameters to check whether $\dot{\nu}$ follows the predicted power-law model, and comment on the result. **[6]**

3. One of the events detected by LIGO was produced by the coalescence of a binary black hole of component masses $m_1 = 85 M_\odot$ and $m_2 = 65 M_\odot$, respectively. Throughout this problem work at the leading Newtonian order.

The binary's total mass, M , reduced mass, μ , and chirp mass, $\mathcal{M}$, are related to m_1 and m_2 according to the following expressions:

$$M = m_1 + m_2, \quad \mu = m_1 m_2 / M, \quad \mathcal{M}^{5/3} = \mu M^{2/3}.$$

- (a) Estimate the orbital separation of the binary when the signal entered the LIGO sensitivity band. [5]
- (b) Estimate the maximum gravitational wave frequency produced by the merger of this binary. Clearly state your assumptions. [5]

- (c) The rate at which energy is lost to to gravitational waves is:

$$\frac{dE}{dt} = -\frac{32}{5}(\pi f \mathcal{M})^{10/3}, \quad [c = G = 1].$$

Assuming this binary evolved only due to gravitational wave emission, calculate the maximum orbital period at which the binary could have formed to be observed by LIGO. [11]

- (d) Hence, compute the total energy that was radiated away in gravitational waves during the entire binary lifetime. [9]

Formula Sheet

Required module specific formulae are given in the exam questions themselves.

Astrophysical Constants and Units

Astronomical Unit	AU	$1.50 \times 10^{11} \text{ m}$
Parsec	pc	$3.1 \times 10^{16} \text{ m}$
Tropical year	y	365.242 mean solar days
Solar luminosity	$L_{\odot}$	$3.84 \times 10^{26} \text{ W}$
Absolute bolometric magnitude of the Sun	M_{bol}	$+4^{\text{m}}75$
Bolometric correction for the Sun	BC	$-0^{\text{m}}08$
Apparent visual magnitude of the Sun	$m_v(\odot)$	$-26^{\text{m}}74$
Solar constant	f	$1.36 \times 10^3 \text{ W m}^{-2}$
Solar mass	$M_{\odot}$	$1.989 \times 10^{30} \text{ kg}$
Wien constant	b	$2.898 \times 10^{-3} \text{ m K}$
Hubble constant	H_o	$70 \text{ km s}^{-1} \text{ Mpc}^{-1}$
Solar radius	$R_{\odot}$	$6.95 \times 10^8 \text{ m}$
Distance of the Sun from galactic centre		8.3 kpc
Earth mass	$M_{\oplus}$	$5.972 \times 10^{24} \text{ kg}$
Earth radius	$R_{\oplus}$	6371 km

Physical Constants and Units

Acceleration due to gravity	g	9.81 m s^{-2}
Gravitational constant	G	$6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Ice point	T_{ice}	273.15 K
Avogadro constant	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
[<i>N.B.</i> 1 mole $\equiv$ 1 <i>gram-molecule</i>]		
Gas constant	R	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23} \text{ J K}^{-1} \equiv 8.62 \times 10^{-5} \text{ eV K}^{-1}$
Stefan constant	σ	$5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Rydberg constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
	$R_\infty hc$	13.606 eV
Planck constant	h	$6.626 \times 10^{-34} \text{ J s} \equiv 4.136 \times 10^{-15} \text{ eV s}$
	$h/2\pi$	$\hbar$ $1.055 \times 10^{-34} \text{ J s} \equiv 6.582 \times 10^{-16} \text{ eV s}$
Speed of light <i>in vacuo</i>	c	$2.998 \times 10^8 \text{ m s}^{-1}$
	$\hbar c$	197.3 MeV fm
Charge of proton	e	$1.602 \times 10^{-19} \text{ C}$
Mass of electron	m_e	$9.109 \times 10^{-31} \text{ kg}$
Rest energy of electron		0.511 MeV
Mass of proton	m_p	$1.673 \times 10^{-27} \text{ kg}$
Rest energy of proton		938.3 MeV
One atomic mass unit	u	$1.66 \times 10^{-27} \text{ kg}$
Atomic mass unit energy equivalent		931.5 MeV
Electric constant	ϵ_0	$8.854 \times 10^{-12} \text{ F m}^{-1}$
Magnetic constant	μ_0	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Bohr magneton	μ_B	$9.274 \times 10^{-24} \text{ A m}^2 (\text{J T}^{-1})$
Nuclear magneton	μ_N	$5.051 \times 10^{-27} \text{ A m}^2 (\text{J T}^{-1})$
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	$7.297 \times 10^{-3} = 1/137.0$
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12} \text{ m}$
Bohr radius	a_0	$5.2918 \times 10^{-11} \text{ m}$
angstrom	$\text{\AA}$	10^{-10} m
barn	b	10^{-28} m^2
torr (mm Hg at 0 °C)	torr	$133.32 \text{ Pa (N m}^{-2}\text{)}$

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.